# Wael Chabchoub

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## Summary

Dedicated and highly skilled Software Engineer with over 2 years of experience in developing robust Java applications and implementing cutting-edge machine learning models. Proficient in designing, developing, and deploying end-to-end software solutions that deliver high-quality user experiences and drive business growth.

## Professional Experience

### Software Engineer

*03.2026 - Present · LIMAA Technologies, Cologne, DE*

Developing LIMAApath, a web-based digital pathology platform built as in-vitro diagnostic medical device software (IVDR, IEC 62304), from image conversion to the web viewer. Designing and maintaining the Java/Spring Boot microservice backend (PostgreSQL, RabbitMQ) and the Python CZI to OME-Zarr conversion pipeline, and building the React/TypeScript viewer with multi-channel display, annotation tools and saved user settings. Owning CI/CD for all services (GitHub Actions, Docker, GHCR, Nexus) with security scans, SBOMs and signed artifacts, performing penetration testing of the web application and its APIs, automating audit evidence for the regulated process, and collaborating with Product, QA and RA on requirements, risk and architecture documentation and release management.

**Skills:** Java · Spring Boot · Python · TypeScript · React · PostgreSQL · RabbitMQ · Docker · GitHub Actions · IEC 62304 · Penetration Testing

### Software Engineer

*03.2023 - 02.2026 · Invitec GmbH & Co. KG, Aachen, DE*

Revolutionized hospital operations by designing and developing an advanced software application that streamlined internal processes, improving patient care delivery and administrative efficiency, minimized the risk of medical errorsresulting in improved patientsatisfaction and an increase in hospital staff productivity

**Skills:** Aurelia · Spring Boot · Jenkins · Docker · Asciidoctor · Java · TypeScript

### Cyber Security Instructor

*06.2021 - 02.2023 · GOMYCODE, Sousse, TN*

Designed and led hands-on laboratory workshops that engaged up to 200 students at a time implemented practical exercises which resulted in significant skill enhancement as evidenced by improved projectcompletion rates among participants.

**Skills:** Penetration Testing · Scripting · Intrusion Detection · Intrusion Prevention · Malware Analysis

### Artificial Intelligence Instructor

*12.2020 - 02.2023 · GOMYCODE, Sousse, TN*

Developed and conducted hands-on AI-focused laboratory workshops, engaging studentsin practical exercisesto build skillsin machine learning, data analysis, and algorithm development, resulting in improved comprehension and application of AIconcepts.

**Skills:** Supervised and Unsupervised algorithms · Data Preprocessing · Features Extraction · Classification · Regression · Clustering · NLP

### Internship

*06.2022 - 07.2022 · ETC Tunisie, Remote*

Redesigned the car rental process by creating an intuitive website with integrated admin and user modules,significantly enhancing user experience and improving operational efficiency.

**Skills:** HTML5 · CSS3 · JSON Web Token (JWT) · Angular · Symfony · REST APIs · API Platform · TypeScript · PHP · Bootstrap

### Internship

*08.2021 - 09.2021 · Gpro consulting, Sousse, TN*

Streamlined the student recognition process at the university by designing and implementing a mobile application featuring advanced face detection and recognition capabilities, demonstrating expertise in innovative app development and image processing.

**Skills:** Flutter · Flutter Mobile Vision

### Internship

*07.2021 - 08.2021 · Inside Conseils, Sousse, TN*

Developed an Odoo module to optimize the university's internal processes,significantly improving the efficiency and accuracy of administrative operations.

**Skills:** Python · Odoo

## Education

### Master Degree in Software Engineering

*2020 - 2023 · Higher Institute of Applied Sciences & Technology, Sousse, Tunisia*

Advanced studies focusing on software architecture, scalable system design, and emerging technologies in software engineering.

### Bachelor of Computer Science

*2017 - 2020 · Higher Institute of Applied Sciences & Technology, Sousse, Tunisia*

Comprehensive foundation in computer science principles, including data structures, algorithms, and object-oriented programming.